

Exp No. 4: OFDMA Resource Block Allocation

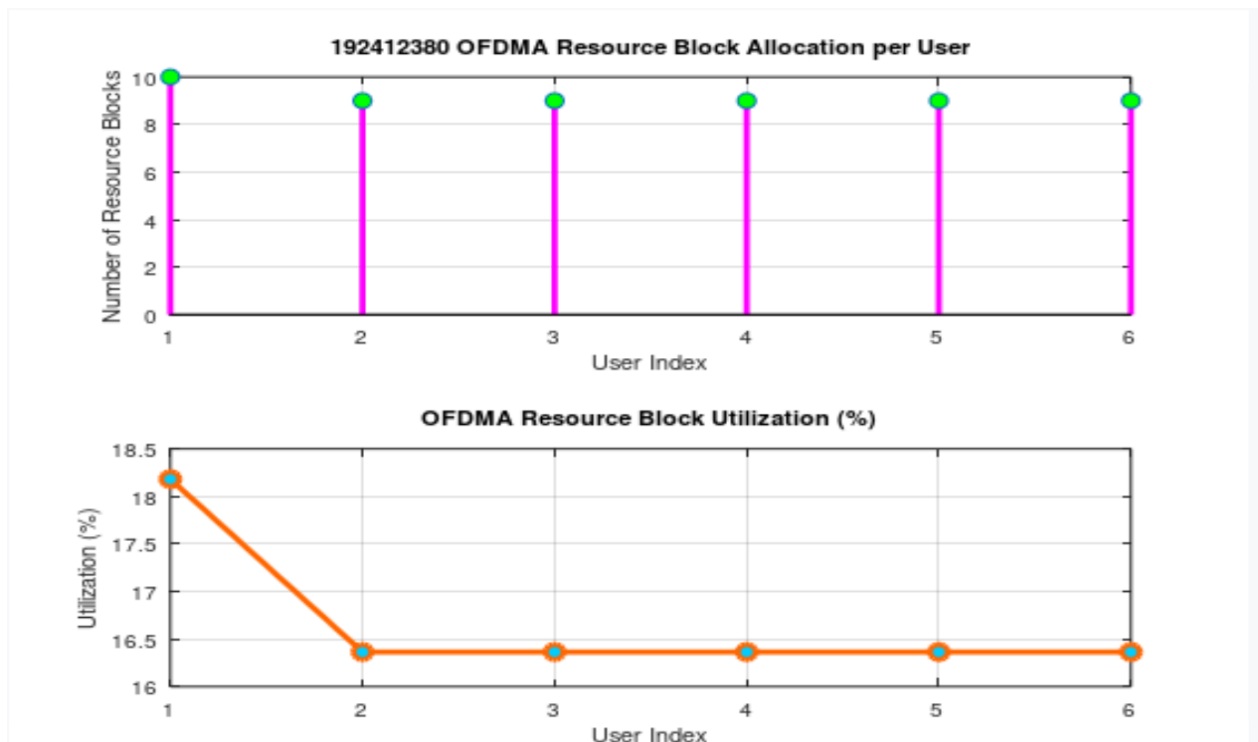
AIM: To simulate and analyze OFDMA (Orthogonal Frequency Division Multiple Access) resource block allocation and evaluate system performance in terms of resource utilization and user throughput using MATLAB.

Objective 1: To determine the Number of Resource Blocks (RBs) allocated in an OFDMA system based on available bandwidth and system configuration.

Program :

```
1  clc;
2  clear;
3  close all;
4
5  % System Parameters
6  bandwidth = 10e6;      % 10 MHz
7  rb_size = 180e3;       % 180 kHz per Resource Block
8
9  % Total Resource Blocks
10 total_RB = floor(bandwidth / rb_size);
11
12 % Users
13 users = 1:6;
14
15 % Equal RB Allocation
16 base = floor(total_RB / length(users));
17 alloc = base * ones(1, length(users));
18
19 % Distribute Remaining RBs
20 alloc(1:mod(total_RB, length(users))) = ...
21     alloc(1:mod(total_RB, length(users))) + 1;
22
23 % Resource Utilization
24 utilization = (alloc / total_RB) * 100;
25
26 % Display Results
27 fprintf('Total Resource Blocks = %d\n\n', total_RB);
28
29 fprintf('User\tRB Allocation\tUtilization (%%)\n');
30 fprintf('-----\n');
31
32 for i = 1:length(users)
33     fprintf('%d\t%d\t%.2f\n', users(i), alloc(i), utilization(i));
34 end
```

OUTPUT :



Objective 2: To evaluate the **Resource Block Utilization (%)** by analyzing how efficiently available RBs are assigned to active users.

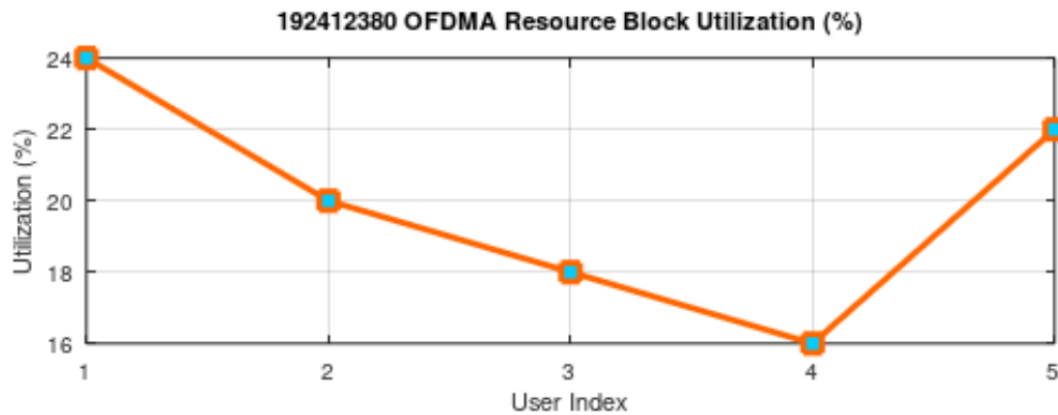
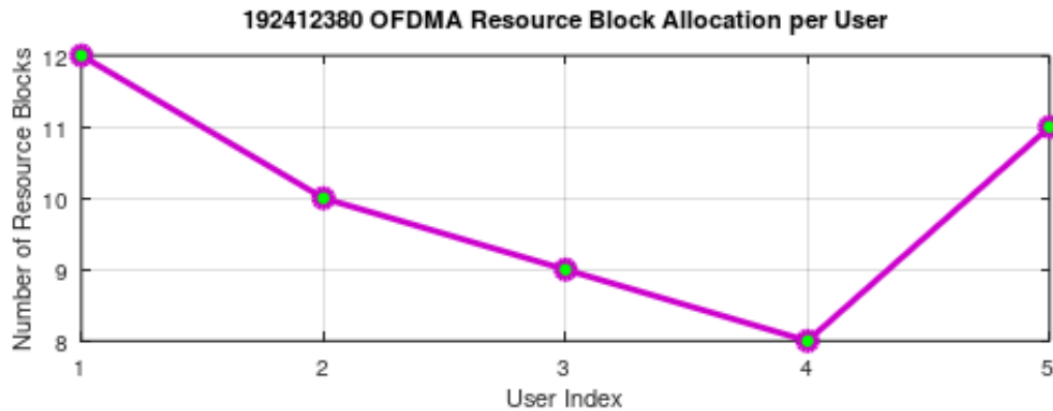
Program

```

1  clc;
2  clear;
3  close all;
4
5  % Total System Resource Blocks
6  total_RB = 50;
7
8  % Active Users
9  users = 1:5;
10
11 % Resource Block Allocation
12 alloc = [12 10 9 8 11];
13
14 % Resource Block Utilization (%)
15 util = (alloc / total_RB) * 100;
16
17 % Display Results
18 fprintf('User\tRB Allocation\tUtilization (%)\\n');
19 fprintf('-----\\n');
20
21 for i = 1:length(users)
22     fprintf('%d\t%d\t%.2f\\n', users(i), alloc(i), util(i));
23 end
24
25 figure;
26
27 %----- Graph 1 -----%
28 subplot(2,1,1);
29
30 plot(users, alloc, ...
31     '-o', ...
32     'color', [0.8 0 0.8], ...      % Purple line
33     'MarkerFaceColor', [0 1 0], ...% Green markers
34     'LineWidth', 2);

```

OUTPUT :

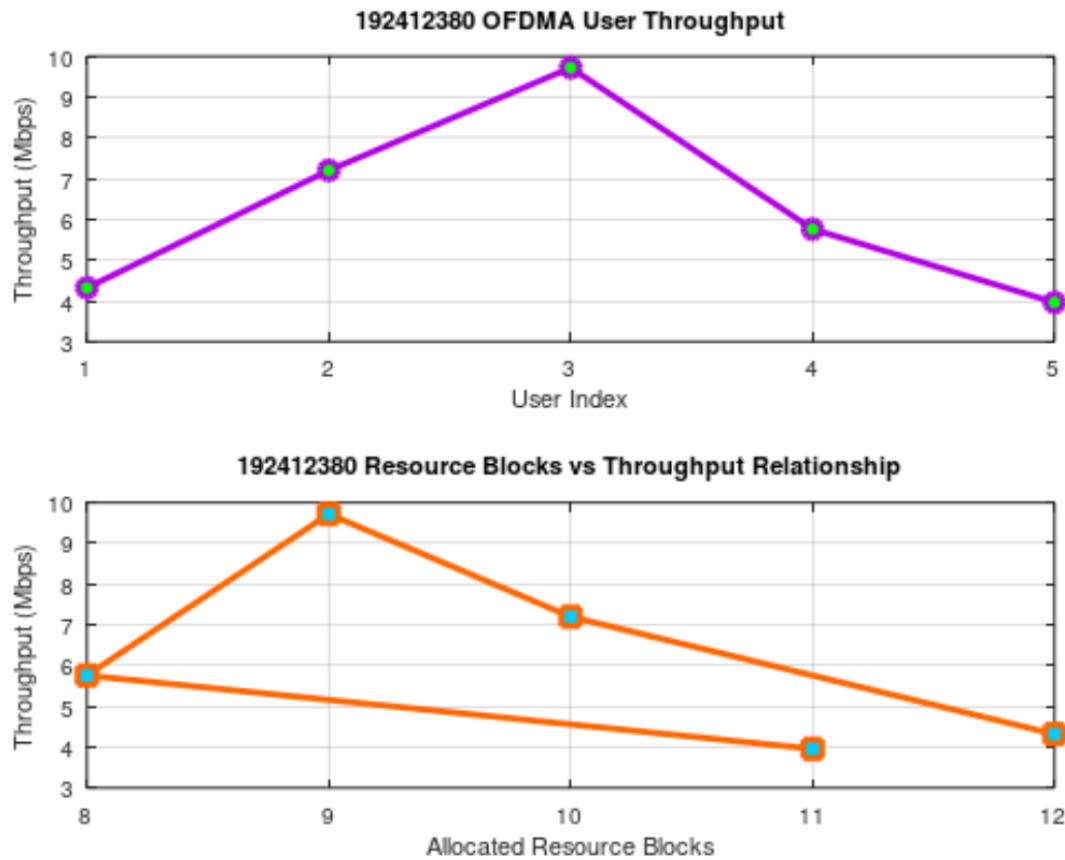


Objective 3: To calculate the User Throughput (Mbps) based on allocated resource blocks and modulation efficiency in the OFDMA system.

program

```
1  clc;
2  clear;
3  close all;
4
5  % System Parameters
6  rb_bw = 180e3;           % Bandwidth per Resource Block (180 kHz)
7
8  % Users
9  users = 1:5;
10
11 % Allocated Resource Blocks
12 alloc_RB = [12 10 9 8 11];
13
14 % Modulation Efficiency (bits/symbol)
15 % User1 = QPSK, User2 = 16QAM, User3 = 64QAM, etc.
16 mod_eff = [2 4 6 4 2];
17
18 % Throughput Calculation (Mbps)
19 throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;
20
21 % Display Results
22 fprintf('User\tAllocated RB\tThroughput (Mbps)\n');
23 fprintf('-----\n');
24
25 for i = 1:length(users)
26     fprintf('%d\t%d\t%.2f\n', users(i), alloc_RB(i), throughput(i));
27 end
28
29 figure;
30
31 %----- Graph 1 -----%
32 subplot(2,1,1);
33
34 plot(users, throughput, ...
```

OUTPUT :



Result:

- 1.The propagation loss was successfully computed for Sub-1 GHz, Mid-band, and mmWave frequency bands using the Free Space Path Loss (FSPL) model.
- 2.It was observed that propagation loss increases with increase in operating frequency, with mmWave showing the highest attenuation and Sub-1 GHz showing the lowest loss.
- 3.The comparative analysis of spectrum bands clearly demonstrates that lower frequency bands are more suitable for long-range communication, while higher frequency bands require advanced techniques like beamforming for efficient performance.